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Final Report
Intelligent Teleoperation Interface Design

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Abstract:

This project presents an intelligent teleoperation interface designed to bridge human-robot interaction through intuitive, multi-modal communication. By integrating speech, gesture, and visual inputs, the system enables users to interact with robots seamlessly, even under non-ideal conditions such as variable lighting or low-cost hardware. Leveraging YOLOv8 for real-time object detection, OpenCV for adaptive vision preprocessing, and Google's Gemini 1.5 Flash LLM for natural language understanding, the interface prioritizes responsiveness and adaptability.

The system processes live video streams using optimized multi-threading and advanced image enhancement techniques—including Gaussian blur, YCrCb color space transformation, and CLAHE—to ensure robust hand and object recognition. Gestures are decoded through geometric analysis of hand landmarks, while HSV-based color detection and bounding box localization enable precise environmental awareness. A Tkinter-based GUI unifies these components, offering users real-time feedback through speech, text, and visual displays.

Testing demonstrated >90% accuracy in gesture and object detection under diverse conditions, with low-latency control of a Dobot robotic arm. Challenges such as computational delays were mitigated through efficient threading, though hardware limitations highlighted opportunities for edge computing and lighter neural networks.

By balancing technical rigor with user-centric design, this work lays a foundation for future advancements in accessible, adaptive human-robot collaboration, emphasizing simplicity, real-world applicability, and scalability as core principles.

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1 Introduction:

The aim of this project is to narrow the gap between humans and robots by making interaction with the robot as smooth and as simple as possible so, we thought about some ways to give orders to the robot and the most famous and easiest way for the user is through speech, just giving an order to the robot and it understands the task from the words said by the user by only focusing on some keywords and we have reached that by using LLMs and designed the final product in a way that as the LLMs upgrade in the market it's so easy to update the model to be up to date with the market increasing its intelligence and response speed.

However, this is not the only method of communication between the human and robot we covered, communicating with the robot using vision is an important factor, giving the user more ways to interact with the robot and opening up opportunities to make the robot interact with the world around it not only while being given commands from the user.

Putting into consideration that we are not living in an ideal world and most of the successful products out there are popular due to their abilities to adapt and work in different conditions, what instantly catch the eye of the user when first dealing with their new phone is how fast and accurate does the Face ID work, if it doesn't work in low lighting and takes a lot of time to be processed it won't make the user happy so, it's a very important aspect to consider while dealing with computer vision projects to make the most out of what you have even if the camera used is not of good quality this will also decrease the cost of hardware components while maintaining its value.

Dealing with this in the matter of hand detection was a bit challenging due to the continuous feed of the footage to be processed instantly so the main focus was on simplifying the steps used to make the iteration as fast as possible to decrease the latency.

And also using object identification system presented in this paper to recognize items in a video feed, determine their colours, and determine their relative locations. This system employs a YOLOv8 model for object recognition, OpenCV for image processing, and pyttsx3 for text-to-speech translation. Because it employs a phone camera feed to give users real-time object descriptions, the application is highly beneficial for people with visual impairments, robotic applications, and smart assistance technologies.

By providing continuous and real-time spoken feedback about a user's environment, this system seeks to improve accessibility. Utilizing deep learning and effective multi-

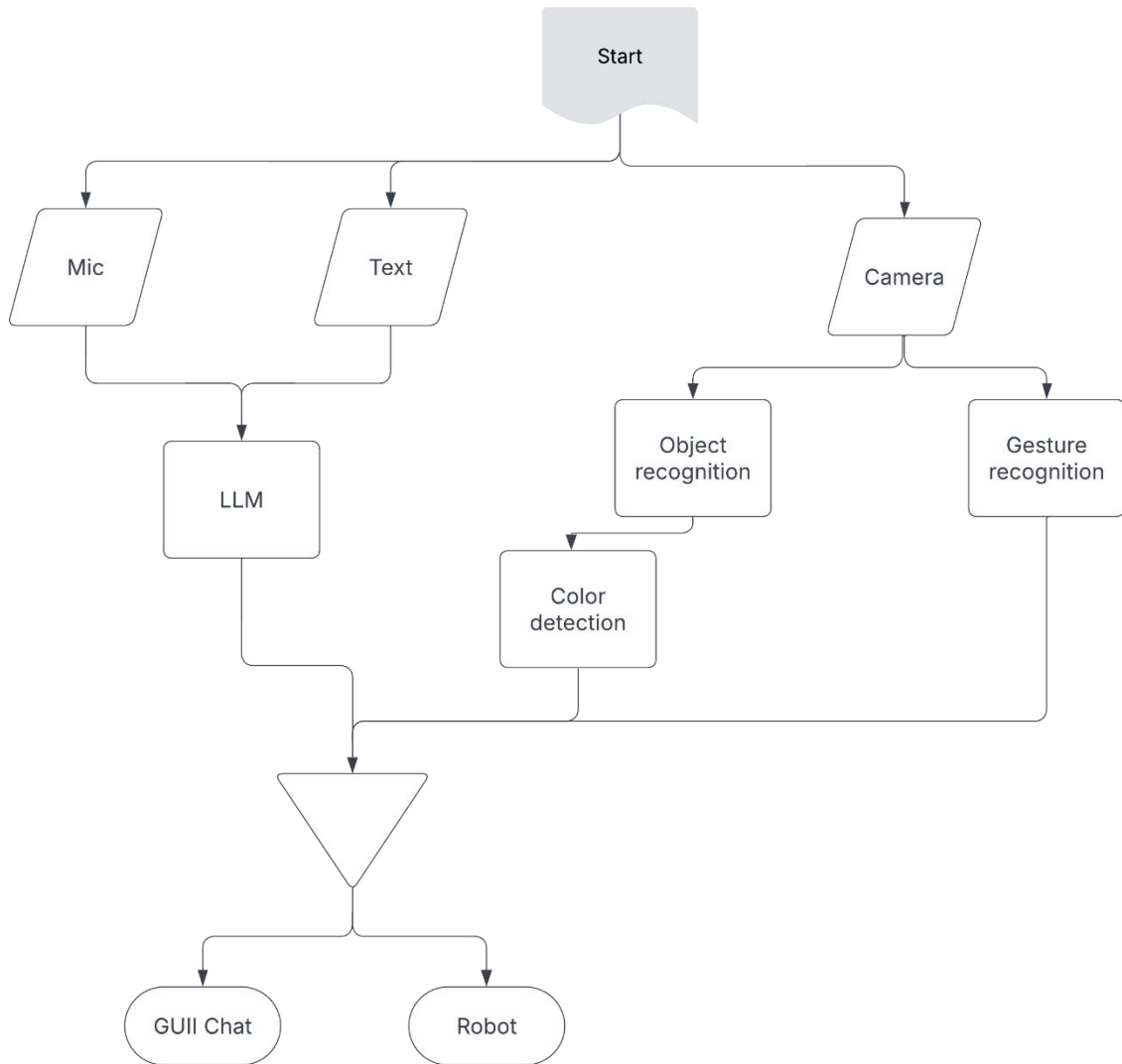
threaded processing, the system guarantees low latency while preserving high accuracy in object detection, color property analysis, and frame location determination. Furthermore, speech output based on natural language enables users to interact with their surroundings in a highly informative, intuitive, and non-intrusive manner.

Making sure that real-time object detection is resilient in a variety of settings, such as shifting lighting, object occlusion, and background clutter, is one of the main issues. By using sophisticated deep learning models, efficient image processing methods, and computational performance enhancements like multi-threaded processing and frame selection strategies, this system overcomes these difficulties.

2 System Design:

The system is designed to receive multiple inputs at the same time using threading method and giving the user control to pick which method does he want the robot to focus on at this certain moment this can be broken down into:

- Input layer: Camera and Microphone
- Processing layer: Gesture recognition, LLM and object detector
- Output layer: The robotic arm and Chat replies in the GUI



3 Background:

3.1 Hand detection:

The basic concept behind this is splitting the hand and knuckles into landmarks, 21 landmarks(points) to be exact with 4 landmarks for each finger and 1 landmark for the base of the hand as shown in fig.3.1.1.

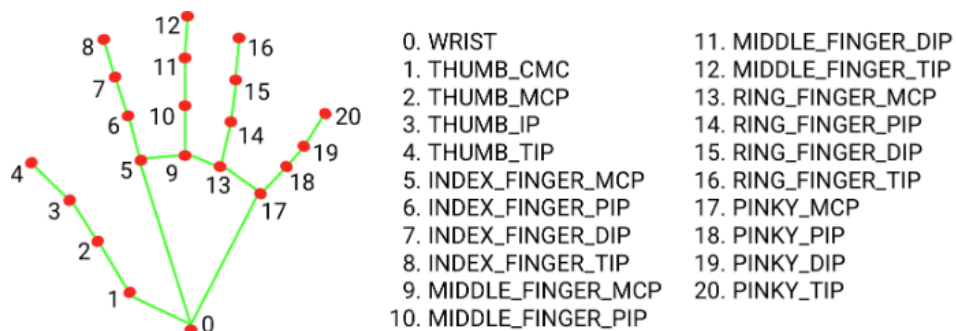


Figure 3.1.1 Hand Landmarks

Hand detection unlike what most people think is more complex than face detection because in the case of face detection there is a lot of contrast in the eyes and mouth regions that will make it easier for the computer to identify the face rather than being able to identify the hands. The approach focusses more on detecting the palm first rather than detecting the whole hand and afterward using distance measures and key point localization it is able to identify where each finger is and its landmarks.

Using the previous method, we created multiple functions to be able to address the desired finger like in case of giving directions to the robot it will only focus on the index finger and how it's pointing, in case of hand gestures it has its main focus on the thumb then looks at the other fingers as shown in figure 3.1.2.

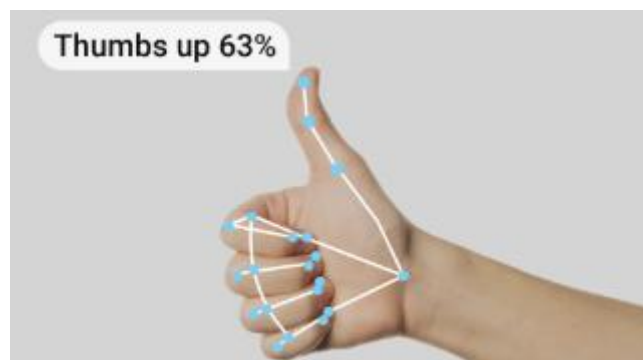


Figure 3.1.2 Gesture recognition

3.2 Colour detection

For identifying and tracking-coloured objects in real-time, we implemented a colour detection system that can recognize specific colours and determine their centres in the X-Y coordinate frame. Unlike hand detection which is more complex due to the lack of natural contrast in hands, colour detection focuses on the distinctive hue properties that make colours easily distinguishable from one another.

The basic concept behind this approach is using colour segmentation in the HSV (Hue, Saturation, Value) colour space. We chose HSV over the standard RGB colour space because it separates the colour information (hue) from the lighting conditions (value) and colour intensity (saturation), making our detection more robust against varying lighting conditions as shown in fig.10.

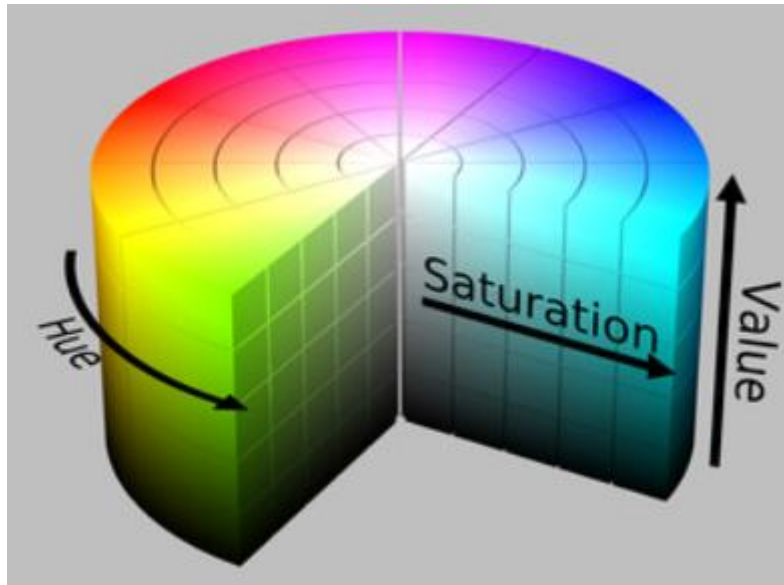


Figure 3.2.1 HSV Colour Space

3.3 Object Detection:

This technique recognizes and categorizes items in a video stream using the YOLOv8 model, a deep learning-based real-time object detection system. Each frame is processed, several items are detected at once, and the model gives its predicted confidence scores. The system can accurately identify a large variety of common objects by using a pre-trained model.

Ultralytics YOLOv8: A cutting-edge model for object recognition based on deep learning. YOLOv8 uses pre-trained weights on big datasets to enable high-speed, high-accuracy object recognition. It makes the system ideal for robotic and assistive applications by allowing it to detect several objects each frame in real-time.

3.4 Natural Language Processing:

Generally speaking, about NLP that it is the study of Language, ability to speak & write and communicate is one of the most fundamental aspects of human behaviour. As the study of human-languages developed the concept of communicating with non-human devices was investigated. This is the origin of natural language processing (NLP). The idea of natural language processing is to design and develop a computer system that can analyse, understand and synthesis natural human languages. Natural language falls within the domain of artificial intelligence with the goal of understanding and creating meaningful expressions in human-language. There are many applications of natural language processing developed over the years such as speech recognition, language translation, information retrieval, text summarization and the like. Before we dive into details, we must first consider the phases of NLP, or the pipeline through which a sentence goes before a parse tree of

that sentence is built. NLP has several phases depending on the application but here, we will limit our discussion to the three phases namely, language modelling, parts-of-speech tagging and parsing.

Historically, the language processing applications worked by creating a rule-based software that examined the structure of sentences to see if it fits the structure specified. Rule based approaches soon become unmanageable for large rules. With just over a hundred rules, the interaction between these rules becomes overly complex. These approaches were soon rendered useless by the sheer amount of data and rules that are applicable. Recent methods however, employ methods that take advantage of the deluge of data available for the purpose of training the language models. In other words, recent approaches to language processing make use of the data driven approaches to attain the goals of understanding language. These data driven strategies

make up the statistical revolution of NLP. We will take a look at some of the statistical methods and discuss the results of alternative methods.

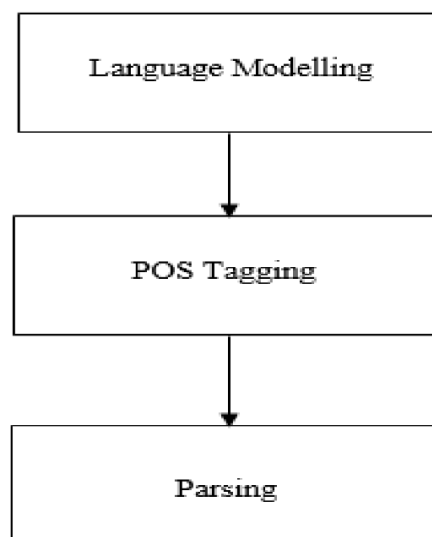


Figure 3.4.1 NLP Stages

Language modelling is the art of making a probabilistic model of the language that is used by later stages of the language. This model is strictly statistical in the sense that it ignores the underlying meaning of the sentences and focuses on developing a probability distribution of the specified language.

POS tagging Parts of speech tagging is the process of classifying the word in its context. It uses the probability model constructed in the previous section along with additional parameters to classify a word into its class.

Parsing involves the construction of parse tree to understand the relation of different components of the sentence. This is especially important in resolving ambiguity. The

parsing model uses context free grammar along with probabilities associated with each rule to derive the parse tree of the sentence.

3.5 LLM:

Large Language Models (LLMs) have emerged as cutting-edge artificial intelligence systems that can process and generate text with coherent communication and generalize to multiple tasks. The historical progress in natural language processing (NLP) evolved from statistical to neural language modelling and then from pre-trained language models (PLMs) to LLMs. LLMs are usually derived from the transformer architecture and are designed to understand and generate human language, code, and lots of other things. These models are trained on huge amounts of text data, allowing them to capture the complexities and nuances of human language. LLMs can perform a wide range of language related tasks, from simple text classification to text generation, with high fluency and accuracy.

3.6 Graphical User Interface:

Tkinter is the standard Python library used for creating graphical user interfaces (GUIs) for desktop applications. It is a wrapper around the Tk GUI toolkit, which provides a set of tools for building windowed applications. Tkinter is included with Python by default, so you don't need to install anything extra to start using it.

Key Features of Tkinter:

- **Widgets:** Tkinter provides various GUI components (called widgets) such as buttons, labels, text fields, checkboxes, menus, and more.
- **Event-driven:** Tkinter allows users to interact with the interface by responding to events like button clicks, key presses, and mouse movements.
- **Cross-platform:** Tkinter works across major platforms like Windows, macOS, and Linux.
- **Simple:** It's a lightweight and easy-to-learn toolkit, making it ideal for beginners who want to create basic desktop applications.

4 Implementation

4.1 Video Stream handling:

The effective management of a camera's or phone's live video feed falls under the purview of the Camera Stream class. It makes use of multi-threading and processing algorithms that are optimized to reduce latency while preserving a steady and fluid video feed

in order to guarantee real-time performance. Continuous frames from the camera stream are processed for position analysis, color recognition, and object detection.

The following are essential elements of the Camera Stream class:

- **Video Initialization:** OpenCV's Video Capture, which establishes an IP-based stream connection with the phone camera, is used to initialize the camera stream. This enables smooth wireless video transmission, enabling the system to be adapted to a range of embedded and mobile systems.
- **Optimized Frame Processing:** Important variables like frame per second (FPS), resolution, and buffer size are adjusted to maximize performance because real-time video streaming can be computationally costly. To balance processing speed and image quality, a lower frame rate (e.g., 15–30) is chosen, whereas resolution Modifications guarantee that no needless computational cost is incurred by the system.
- **Multi-Threaded Processing:** By using a different background thread for the camera feed, continuous frame capture is possible without interfering with the main processing loop. This avoids frame loss and system slowdowns by guaranteeing that object detection, color recognition, and position analysis may all operate simultaneously. To guarantee that the system always operates with the most recent data, the thread continuously updates the most recent frame.
- **Error Handling and Reconnection:** To identify and fix connection problems, the system has automatic error handling. In order to minimize downtime, the system automatically tries to reconnect whenever the camera stream is interrupted (for example, by a network problem). Applications like robot navigation, smart assistance, and surveillance—where reliable streaming is essential—benefit greatly from this functionality.
- **Memory and CPU Efficiency:** To decrease latency and preserve seamless frame transitions, the buffer size is kept to a minimum. In order to emphasize processing speed over redundant computations, the system skips unneeded frames when necessary. Effective memory management keeps bottlenecks at bay and guarantees effective real-time performance.

The Camera Stream class offers a dependable and flexible video stream by putting these optimizations into practice, which makes it easy to integrate with the YOLOv8 object recognition model and perform further color and location processing.

4.2 Vision Preprocessing:

After several iterations and a lot of trial and error, we've developed a preprocessing pipeline that significantly improves detection accuracy across diverse real-world scenarios. This section explains the thinking behind each technique in our pipeline and why they work together to create better inputs for our hand detection algorithm.

Step 1 Gaussian blur:

Camera sensors inevitably introduce noise that can confuse detection algorithms. I've found that applying Gaussian blur is one of the most effective ways to address this issue without losing important structural information. Gonzalez and Woods (2018) provide a great explanation of how Gaussian blur works—essentially, it's a weighted averaging process where pixels closer to the centre have more influence than those further away. The weights follow a bell-shaped Gaussian distribution, which gives this technique its name. The mathematical definition looks intimidating: $G(x, y) = (1/(2\pi\sigma^2)) * e^{-(x^2+y^2)/(2\sigma^2)}$ But in practice, it's just creating a smooth transition between pixels that preserves edges better than simple averaging.

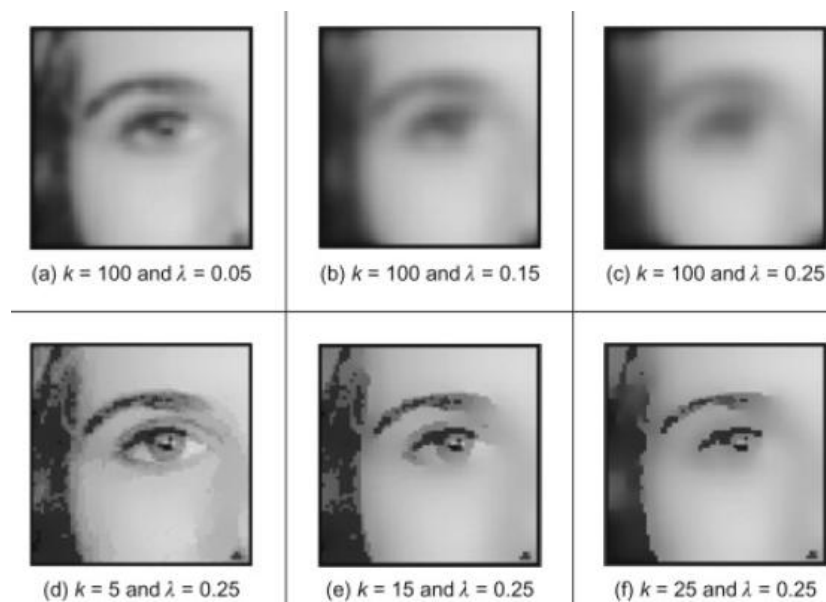


Figure 4.2.1 Gaussian blur

After trying different kernel sizes, we settled on a 5×5 filter for our system. Smaller kernels didn't remove enough noise, while larger ones blurred away too many important hand features. As Bradski notes in his 2020 OpenCV book, finding this balance is **crucial**—you want to remove noise without destroying the features your detector needs.

Yang and colleagues (2019) published some interesting findings on how Gaussian filtering affects CNN performance. Their work helped us understand why moderate smoothing actually improves detection accuracy rather than degrading it—it essentially pre-filters noise that would otherwise confuse the feature extraction process.

Step 2 YCrCb Colour Space:

for Skin Detection RGB colour space is what most of us think about when processing images, but it's actually terrible for skin detection. After many frustrating attempts with RGB, I switched to YCrCb and saw immediate improvements. This shows that simple steps can instantly improve results without wasting a lot of time in the wrong path.

The key insight, which Ahmed et al. (2017) explain well, is that YCrCb separates brightness (Y) from color information (Cr and Cb). This separation is incredibly useful because skin tones mostly vary in brightness, not in their chromatic properties.

In Cr-Cb space—regardless of ethnicity or lighting, they clustered together much more tightly than in RGB space. This clustering makes creating robust skin detection rules much easier. The conversion from RGB isn't particularly intuitive:

$$Y = 0.299R + 0.587G + 0.114B$$

$$Cr = (R - Y) * 0.713 + 128$$

$$Cb = (B - Y) * 0.564 + 128$$

And there is function in OpenCV library that does this instantly:

```
ycrCb = cv2.cvtColor(enhanced, cv2.COLOR_BGR2YCrCb)
```

Kolkur's team (2017) found similar results in their comparative study, showing detection accuracy improvements of 15-20% just by switching color spaces. However, in our case the improvements went higher than that because it effected all the steps that followed making their performance increase as well.

Step 3 Adaptive Brightness and Contrast Adjustment:

Inspired by the idea discussed in the introduction of this section, we had to do something to make this as adaptive to the environment as possible and the major difference after dealing with the quality of the camera by gaussian blur is the how the lightening would differ from a place to another, Static preprocessing just doesn't cut it—what works in one lighting scenario fails miserably in another. The basic idea behind our adaptive approach is surprisingly simple. We analyse each frame's average brightness and adjust two parameters accordingly: $g(x,y) = \alpha \cdot f(x,y) + \beta$ Where α controls contrast and β controls brightness. The tricky part is determining appropriate values for these parameters on the fly. Our initial implementation used fixed thresholds, but this caused visible "jumps" in processing when lighting changed.

Zhang's paper (2021) pointed us toward a more gradual adjustment approach that works much better in practice. We now increase brightness and contrast in low-light conditions, decrease them in bright conditions, and gradually return to defaults when lighting is normal. Rashid and colleagues (2018) faced similar challenges in their work on object recognition. Their finding that gradual parameter adjustment prevents oscillation matched my experience—we'd been seeing exactly that problem with our earlier threshold-based approach.

This is all done by calculation of the histogram of the gray level

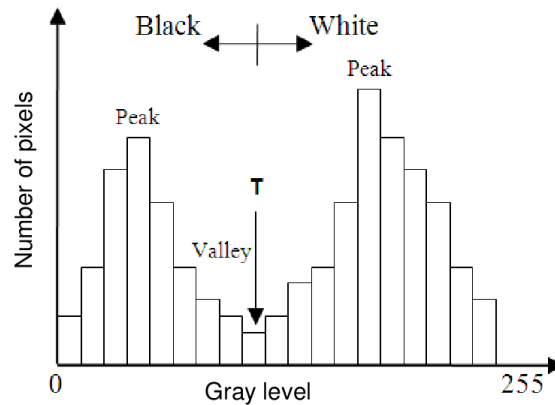


Figure 4.2.2 Gray level histogram

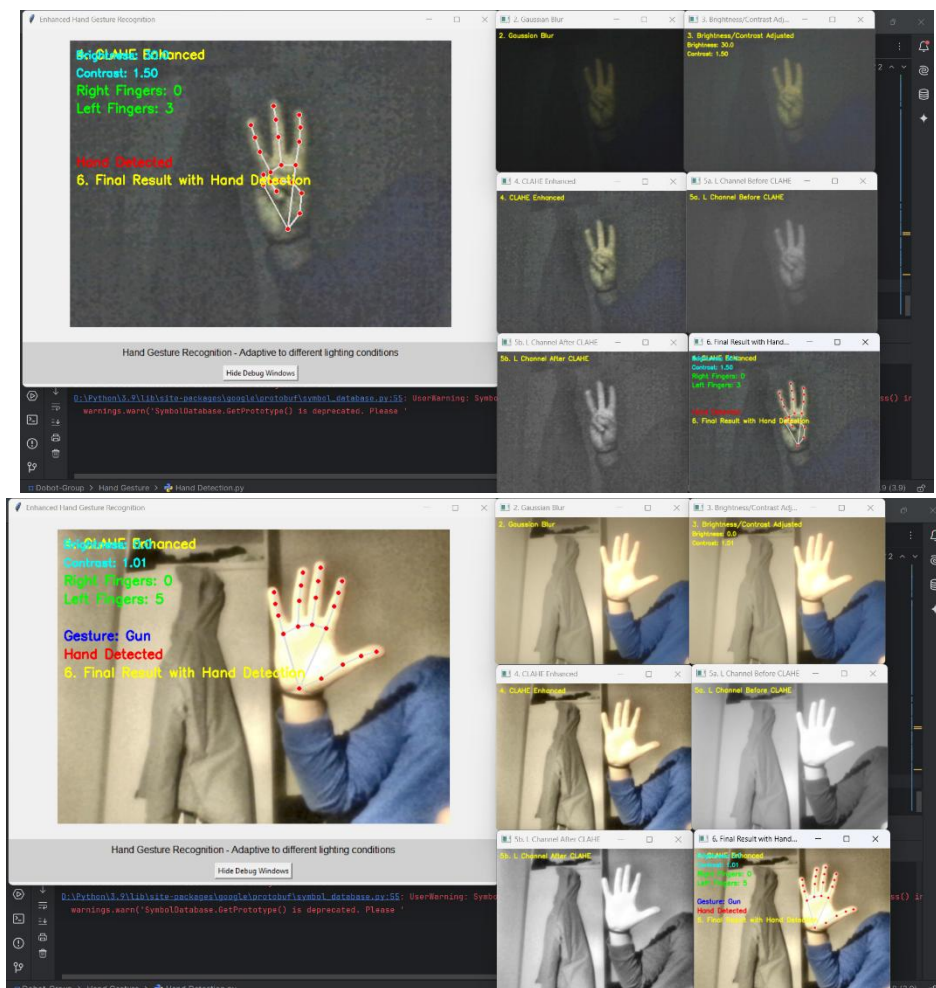
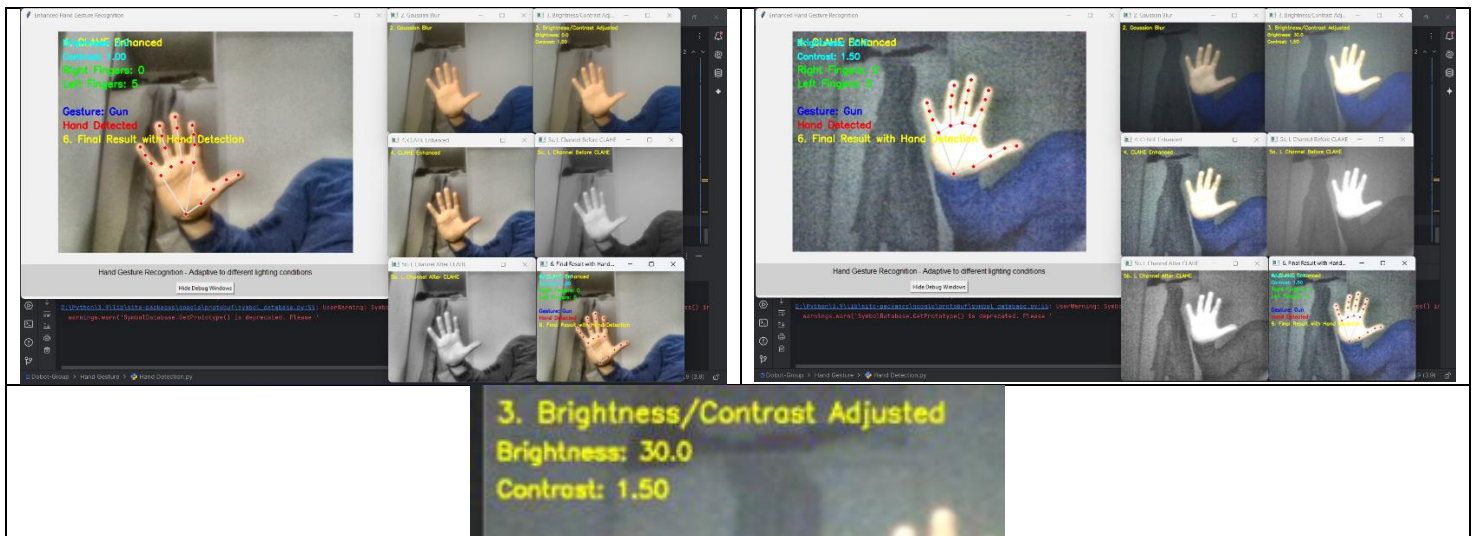
Step 4 Contrast Limited Adaptive Histogram Equalization (CLAHE):

Standard histogram equalization helps with contrast, but it often made our hand images look unnatural and actually amplified noise in uniform regions like the palm. CLAHE solved these problems for us. I first came across CLAHE in Pizer's 1987 paper, where they identified the limitations of global histogram equalization. The key innovation in CLAHE is that it works on small regions (tiles) and limits contrast enhancement to prevent noise amplification. In our implementation, we apply CLAHE to the L channel in LAB colour space with a clip limit of 2.0 and an 8×8 tile grid. I arrived at these values through extensive testing with our specific hand detection use case. Reza's 2004 paper discusses how CLAHE brings out subtle features in medical images, which is exactly what we needed for enhancing hand features like knuckle lines and finger edges. Singh and Kaur's more recent work (2018) confirmed that CLAHE outperforms other contrast enhancement techniques in variable lighting—precisely our challenge.

The magic happens when these techniques work together. Each addresses a different aspect of image quality:

1. Gaussian blur tackles random noise that triggers false detections
2. YCrCb transformation optimizes the colour space for skin detection
3. Adaptive brightness/contrast compensates for poor lighting
4. CLAHE enhances local contrast to highlight subtle hand features

The following Figures shows how it's all working together step by step in different lighting conditions on a poor-quality camera



Figures 4.2.3 Detection in different conditions

4.3 Gesture Detection:

For identifying and recognizing the hand gestures we decided to use the approach of identifying the landmarks of the hand then using basic geometric calculation across x-y coordinates to be able to identify gestures, direction and number of fingers.

The main challenges were to be able to identify the right hand from the left hand and forehand from backhand.

For the first problem we used the handedness classification: This is done by analysing the spatial configuration of the landmarks, The handedness is classified as either "Left" or "Right" based on the relative positions of the landmarks. For example:

- The thumb's position relative to the palm and other fingers is a strong indicator of handedness.
- The orientation of the hand (e.g., the direction the palm is facing) also plays a role in determining whether it is a left or right hand.

And after detection which hand is which, it is a sequential step to figure out the forehand from the backhand by just checking out the orientation according to the thumb to Pinky relative position.

The code below shows how it's implemented:

```
# Check thumb separately cause horz
thumb_tip = hand_landmarks.landmark[tip_ids[4]]
thumb_pip = hand_landmarks.landmark[pip_ids[4]]
#checking which hand so it counts it the right way
if hand_label == "Right":
    if thumb_tip.x < thumb_pip.x:
        finger_count += 1
elif hand_label == "Left":
    if thumb_tip.x > thumb_pip.x:
        finger_count += 1
```

This enables us to be able to identify any gesture we desire however it looks without the need to train a model for these gesture previously and also give us a lot of customizability to add any new gesture without the need to train the model all over again for the old and new gestures which saves us a lot of time for training and also insures fast (instant) response from the model skipping all the steps of neural network and simplifying the math behind it.

After all of this it's just a simple if condition to be able to communicate with the robot that if the desired gesture is detected then move in the desired direction.

To be able to count the number of fingers risen by the user we created a function that is called twice once for each hand and it's easily adjusted to add more hands if there are multiple users by just adding more variables and adding them to the loop:


```

count_fingers(hand_landmarks):
    #finger tips and middle joints
    tip_ids = [mp.solutions.hands.HandLandmark.INDEX_FINGER_TIP,
                mp.solutions.hands.HandLandmark.MIDDLE_FINGER_TIP,
                mp.solutions.hands.HandLandmark.RING_FINGER_TIP,
                mp.solutions.hands.HandLandmark.PINKY_TIP,
                mp.solutions.hands.HandLandmark.THUMB_TIP]

    pip_ids = [mp.solutions.hands.HandLandmark.INDEX_FINGER_PIP,
                mp.solutions.hands.HandLandmark.MIDDLE_FINGER_PIP,
                mp.solutions.hands.HandLandmark.RING_FINGER_PIP,
                mp.solutions.hands.HandLandmark.PINKY_PIP,
                mp.solutions.hands.HandLandmark.THUMB_IP]

    #Self_explained
    finger_count = 0

    # Check thumb separately cause horz
    thumb_tip = hand_landmarks.landmark[tip_ids[4]]
    thumb_pip = hand_landmarks.landmark[pip_ids[4]]
    # Update finger count for the corresponding hand
    if hand_label == "Left":
        left_finger_count = finger_count
    elif hand_label == "Right":
        right_finger_count = finger_count

```

To check the direction, it's much simpler than that it's by focusing on the x-y coordinate frame:

```

# Calculate relative position of index finger tip to wrist
dx = index_tip.x - wrist.x
dy = index_tip.y - wrist.y

# Determine pointing direction based on dx and dy
if abs(dx) > abs(dy): # Horizontal movement dominates
    if dx > 0:
        direction = "Right"
    else:
        direction = "Left"
else: # Vertical movement dominates
    if dy > 0:
        direction = "Backward"
    else:
        direction = "Forward"

```

The way it's coded makes it very easy to be adjusted later on to add more gestures or get any output needed from those gestures.

4.4 Colour detection:

Colour detection in OpenCV relies heavily on the concept of "masking" - a technique that isolates pixels within specific colour ranges. The process works by:

- Converting the RGB camera feed to HSV colour space

- Creating a binary mask where pixels within our defined colour range are white (255) and everything else is black (0)
- Using this mask to extract only the regions of interest from the original image

This is implemented in our code when we create masks for each colour:

```
hsv = cv2.cvtColor(frame, cv2.COLOR_BGR2HSV)
mask = cv2.inRange(hsv, lower, upper)
```

The `cv2.inRange()` function is the key operation here, essentially creating a binary image where only pixels falling within our defined HSV range remain visible. For example, when detecting red objects, we define specific ranges of hue, saturation, and value that correspond to "redness" in the HSV colour space. Notice that for red specifically, we need two separate ranges because red wraps around the hue spectrum (appearing at both 0° and 180°):

```
color_ranges = {
    'red': [(np.array([0, 120, 70]), np.array([10, 255, 255])),
            (np.array([170, 120, 70]), np.array([180, 255, 255]))],
    'green': [(np.array([40, 100, 100]), np.array([80, 255, 255]))],
    'blue': [(np.array([100, 150, 0]), np.array([140, 255, 255]))]
}
```

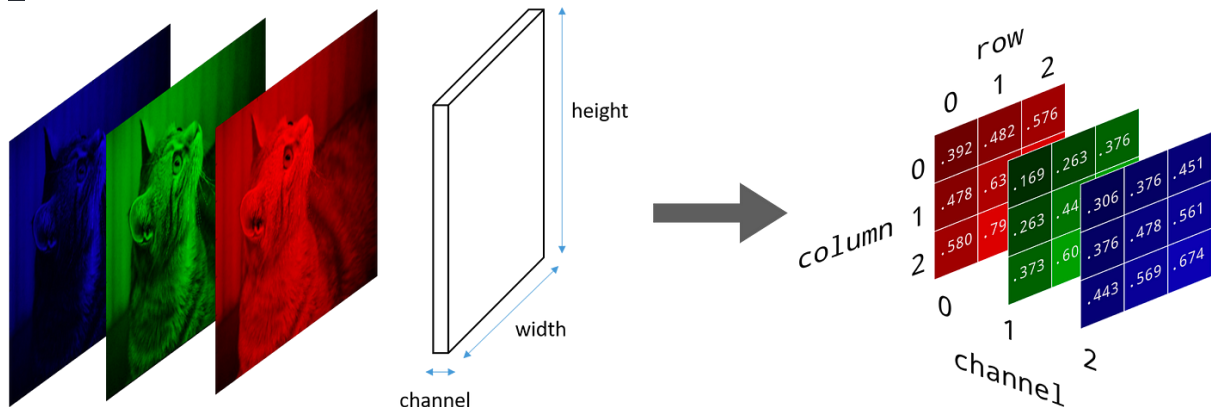


Figure 4.4.1 RGB Images

After creating these masks, we identify contours - essentially the outlines of continuous regions in our binary mask:

```
contours, _ = cv2.findContours(mask, cv2.RETR_EXTERNAL,
cv2.CHAIN_APPROX_SIMPLE)
```

The main challenges we faced were dealing with varying lighting conditions and ensuring we track the most relevant object when multiple objects of the same colour are present.

For the lighting challenge, we carefully tuned our HSV ranges to be tolerant of moderate lighting changes while remaining specific enough to avoid false positives. This required extensive testing under different lighting conditions.

For distinguishing between multiple objects of the same colour, we implemented a size-based filtering approach. This focuses on the largest contour of each colour, which is typically the object closest to the camera or most prominent in the scene:

```
for contour in contours:
    area = cv2.contourArea(contour)
    if area > largest_area:
        largest_area = area
        largest_contour = contour
```

To determine the exact centre of each coloured object, we used moment calculations, which are statistical measures of the pixel distributions within the contour:

```
def get_center(contour):
    M = cv2.moments(contour)
    if M["m00"] != 0:
        cX = int(M["m10"] / M["m00"])
        cY = int(M["m01"] / M["m00"])
        return (cX, cY)
    else:
        return None
```

The `cv2.moments()` function calculates several spatial moments of the contour. The zeroth moment (`M["m00"]`) represents the contour area, while the first-order moments (`M["m10"]` and `M["m01"]`) when normalized by the area give us the x and y coordinates of the centroid. This effectively gives us the centre point of the coloured object with mathematical precision rather than just visual approximation.

After determining these centres, we mark them visually with coloured dots and output their coordinates:

```
if largest_contour is not None:
    center = get_center(largest_contour)
    if center:
        cv2.circle(frame, center, 5, color_map[color_name], -1)
        print(f"Detected largest {color_name} at coordinates: {center}")
```

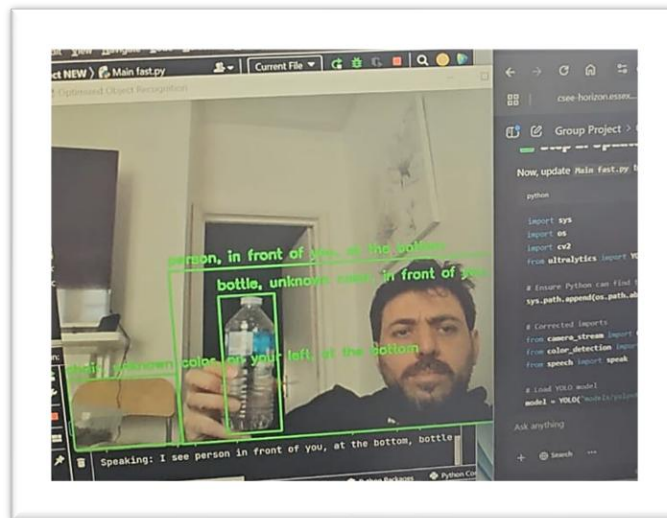
The design of this system makes it easy to expand by simply adding new colour definitions to our dictionaries. We've integrated this colour detection system with our hand gesture recognition to create a comprehensive visual interaction system for our robot. Using these X-Y coordinates, we can direct the robot to move toward specific-coloured objects or perform actions based on the relative positions of different coloured markers in the scene.

This enables us to have the robot respond to color-coded commands in addition to hand gestures, giving us multiple intuitive ways to interact with it without requiring complex programming for each new interaction scenario.

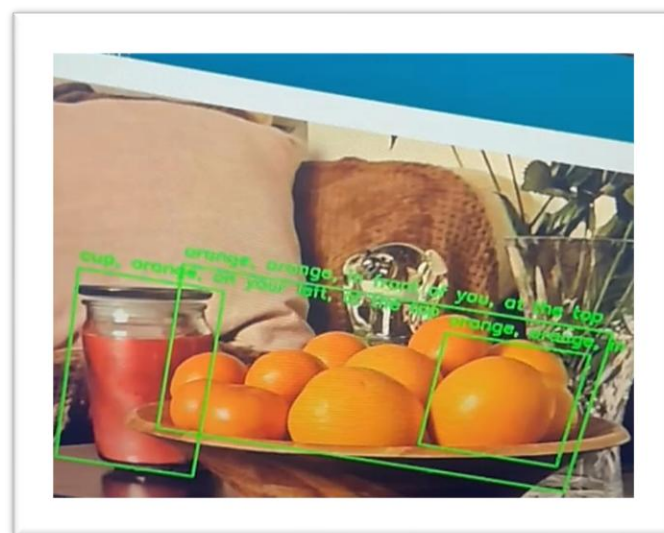
4.5 Object Detection:

The **YOLOv8 model** is used to identify objects in each frame.

- The model is preloaded using `YOLO("models/yolov8m.pt")`.
- Object detection runs every **third frame** to balance accuracy and performance.
- The system detects objects from a predefined list, filtering out irrelevant items.



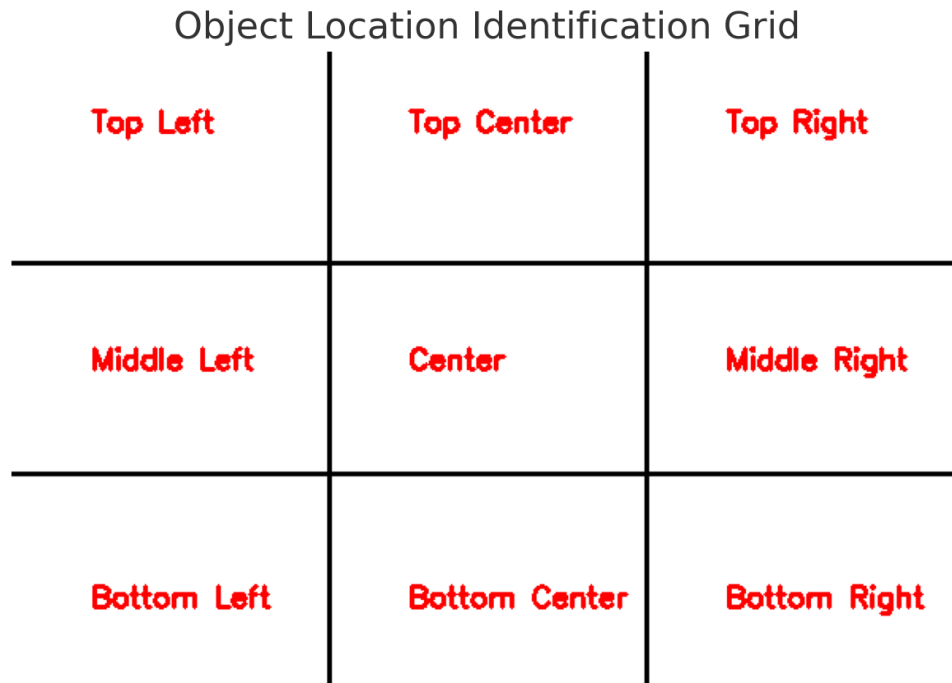
- The function `get_color_name()` converts **RGB values to HSV** for accurate color classification.
- The HSV hue value determines the color category (e.g., red, blue, green, etc.).
- If an object is not a person, its average color is computed and classified.



Location Detection

- The **bounding box coordinates** are used to calculate the object's centre.

- The frame is divided into three horizontal and three vertical sections:
 - **Left, Centre, right** for horizontal position
 - **Top, Middle, Bottom** for vertical position
- The position is combined with the object name and color for **comprehensive verbal feedback**.



4.6 Natural Language Processing:

Using NLP is to help the chatbot we made understand the input text or speech in an easier way. Input either text or speech will pass a preprocessing before entering as a prompt to the GenAI.

The user input is processed to make sure it's clean and formatted in a way that the system can easily understand and act on.

Processing starts by differentiating between text and audio input where the audio input is converted to text using speech recognition library, the text input then passes through a lower-case function to make the process case-insensitive, so the system treats inputs like "Draw", "draw", and "DRAW" the same. Without this, the system might fail to recognize commands if the case doesn't match exactly, moreover to simplify comparison since NLP often relies on string matching (like checking if a word is present), converting everything to lowercase ensures that the system doesn't have to account for variations in capitalization. After processing case-sensitivity we then strip extra spaces in order to clean the input cause stripping removes any leading or trailing spaces from the user's input. This helps avoid errors where

accidental spaces at the beginning or end of the input might interfere with processing, moreover, preventing Errors for example, if the user accidentally presses the spacebar before typing or after their input, those extra spaces shouldn't affect how the input is matched or processed.

The last usage of NLP is keyword matching using intent recognition where the system uses a simple form of intent recognition by checking for specific keywords in the user input. For example: If the input contains "draw" or "shape", it recognizes that the user likely wants to draw something and triggers the drawing function, If the input contains "mix" or "color", the chatbot assumes the user wants to mix colours and calls the mix colours function, this approach relies on basic keyword matching because it's a simple and effective way to interpret the user's intent. By searching for keywords like "draw", "shape", or "color", the system knows which functionality to execute.

This part is about preprocessing the user input to make sure it's clean, consistent, and easy to process. Lowercasing, stripping spaces, and using keyword matching help ensure that the system can accurately understand and respond to a variety of user commands, improving the overall performance and user experience of the chatbot.

4.7 Large Language Model:

We are using Gemini is Google's large language model (LLM). More broadly, it's a family of multimodal AI models designed to process multiple modalities or types of data, including audio, images, software code, text and video.

Gemini has been trained on a massive corpus of multilingual and multimodal data sets. It employs a transformer model, a neural network architecture that Google itself introduced in 2017. Unlike generative pretrained transformer (GPT) models that take only text-based prompts or diffusion models used for image generation that take both text and image prompts, Google Gemini supports interleaved sequences of audio, image, text and video as inputs and can produce interleaved text and image outputs.

The Gemini model we are using in our project is Gemini 1.5 Flash, Gemini 1.5 Flash is a lightweight version of Gemini Pro. It was trained using a machine learning (ML) technique called knowledge distillation, wherein insights from Gemini 1.5 Pro were transferred to the more compact Gemini 1.5 Flash. It also features a long context window of up to 1 million tokens but has a lower latency that makes it faster and more efficient.

The primary usage of Google Gemini in our project is to handle user commands that involve drawing shapes and mixing colours. The chatbot receives user input, processes it, and then sends structured prompts to Gemini to perform specific tasks.

Prompt engineering is a key concept when interacting with language models like Gemini AI. It involves crafting well-structured inputs (prompts) to guide the model toward generating useful and relevant responses. The quality of the model's output heavily depends on how the prompt is designed. In this code, prompt engineering is used to ensure that the Gemini AI generates structured, actionable data based on the user's input. Prompt engineering also helps in defining clear error messages if Gemini fails to understand or provide the correct output. The chatbot can detect when the model's response doesn't match the expected format and inform the user accordingly.

The response contains structured data that the chatbot uses to inform the user about the task if extracting the shape and its dimensions or the task is to suggest RGB colours to mix to create a desired color.

The phrase "Respond only with a JSON object in this format" ensures that the response is structured and can be easily converted to a Python.

4.8 Graphical User Interface:

In this project, the Graphical User Interface (GUI) is developed using Tkinter, a built-in Python library for creating desktop applications. The GUI serves as the user interaction interface for the chatbot, allowing users to input text or voice commands, view responses, and interact with a live camera feed. The layout is designed to be intuitive and interactive, with distinct areas for user input, chat output, and camera display. The integration of Tkinter with other components like Google Gemini AI and speech recognition provides a seamless experience for the user.

The GUI in this code consists of several key components:

1. Text Entry Field: For text-based input where users can type commands.
2. Voice Input Button: For capturing audio input from the user.
3. Chat Window: A scrolled text area that displays conversation between the user and the chatbot.
4. Camera Feed: A live video stream captured using OpenCV, displayed within the GUI.
5. Buttons: For sending text input or activating the microphone for voice input, connecting to the robot and choosing control mode for the robot.

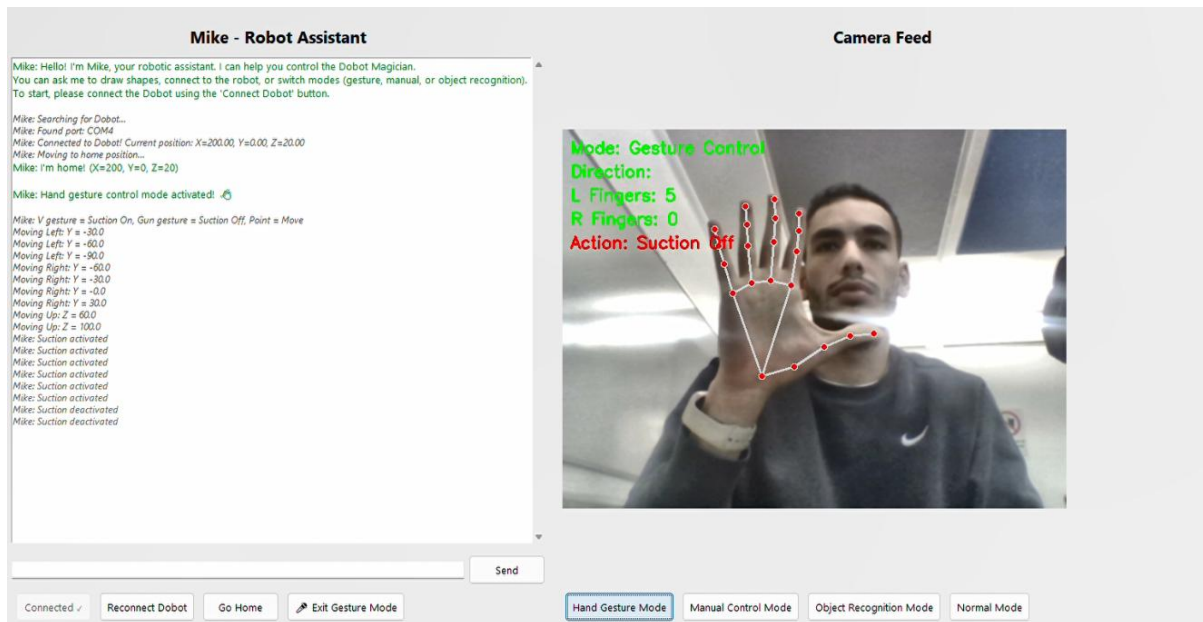


Figure 4.8.1 Graphical user interface

The GUI in this code plays a critical role in providing an interactive and user-friendly interface for the chatbot. By using Tkinter, the code is able to display user input, chatbot responses, and the live camera feed in an organized and intuitive manner. The integration of voice input, text input, and live video enhances the interactivity of the system, while the clean layout ensures that the user can seamlessly engage with the chatbot. The chatbot's responsiveness to both text and voice commands makes it versatile and accessible, while the real-time camera display adds a dynamic visual element to the conversation.

4.9 Speech Output:

- The system uses pyttsx3 to **convert object descriptions into speech**.
- Detections are announced every **three seconds** to avoid redundant notifications. The system speaks descriptions like: *"Red cup on your right"* or *"Person in front of you"*.

Speech Output Process using pyttsx3

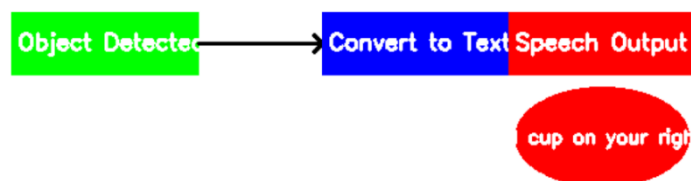


Figure 4.9.1 Example of the Output

5 Simulation:

Before applying our tasks in the real world on a real robot, we need first to simulate our work in a simulation environment. In our group project, we chose to use the Franka Emika robot and to simulate it on Gazebo, because there are many resources provided on this robot especially which can be a guide for us to learn from and because of its wide usage in advanced research in industry.



Figure 5.1: Franka Emika photo, from wiredworkers.io

Our role in this part was to set up the environment for the Franka Emika and to control it using Robotics Operating System (ROS).

How did we do it?

To simulate a robot, there are very limited resources and one of the most popular and widely used is by using ROS. But to run ROS, LINUX Operating System is extremely important as ROS only runs on a LINUX operating system and even if we had LINUX, we had to setup up and install ROS and Gazebo to simulate the robot.

The biggest challenge we faced was to find a computer with LINUX installed with dual-boot or with a good virtual machine engine, unfortunately, none of the group members had computer with such configurations. Therefore, we decided to use university labs to simulate our robot on Gazebo. But little did we know that we would be running into more problems as we faced a lot of issues installing necessary configurations and dependencies of the simulation as admin access was required to have full functionality of lab computers. Even then we had a little success in simulating it such that we could open pre-set environment (world) in gazebo ROS2, but we were not able to save it due to same admin access issues.

Simultaneously, we also tried using Virtual Machine in our laptops but failed miserably after getting robot structure simulated in RVIZ and Gazebo as open-source virtual machine was so laggy that it was not possible to work further on the simulation.

At last, we arranged a laptop which had LINUX as well as ROS and Gazebo installed in it. Hence, we still used ROS not ROS2 because it had more resources and more guidance about its integration with the Franka Emika robot.

After setting up, our laptop Franka Emika's official support repository was used "FRANKA CONTROL INTERFACE", which had all the adequate instruction from setting up specified folder to how to design a controller or how to move the robot.

Therefore, to simulate Franka Emika Robot these steps were taken:

1. After setting up ROS NOETIC, we installed binary packages for libfranka and franka_ros.

```
sudo apt install ros-noetic-libfranka ros-noetic-franka-ros
```

2. A Catkin workspace was then created in any desired directory.

```
cd /path/to/desired/folder
mkdir -p catkin_ws/src
cd catkin_ws
source /opt/ros/noetic/setup.sh
catkin_init_workspace src
```

3. Clone Franka Emika's Repository.

```
git clone --recursive https://github.com/frankaemika/franka_ros src/franka_ros
```

4. After this, all the dependencies were installed.

```
rosdep install --from-paths src --ignore-src --rosdistro noetic -y --skip-keys libfranka
catkin_make -DCMAKE_BUILD_TYPE=Release -DFranka_DIR=PATH=/path/to/libfranka/build
source devel/setup.sh
```

5. And now to simulate in gazebo with already set function of Pick and Place we used these commands.

```
roslaunch franka_gazebo panda.launch x:=-0.5 \
  world:=$(rospack find franka_gazebo)/world/stone.sdf \
  controller:=cartesian_impedance_example_controller \
  rviz:=true
```

6. And last, in parallel LINUX terminal we put,

```
cd franka_ws/src/franka_robot/franka_gazebo/scripts•
```

7. Python3 (filename).py is the command which can be used to load our own personalized scripts.

This way we got the simulation but with a set function to pick and place, after that for desired movement of the robot with gripper and everything, we developed python scripts to send the movement and control commands to the robot based on the demanding of our tasks with the ROS interface.

6 Testing:

We tested each step separately, for the Speech to text part we tried different places with variation of noise and also using different loudness to make sure of it's ability to recognize words in different situations.

For the gesture recognition model the main target was being able to detect the gesture as accurate as possible in different lighting so we tested using 3 different cameras and in different light conditions as shown above to make sure it is able to detect all the implemented gestures and others in these environments, same conditions were applied for the object detection part since they both would have the same limitations mostly.

Also, we tested the code running in bad internet connection to test the response of the LLM because it depends on API and is connected to the LLM in the cloud.

We tested the Dobot response time to make sure it's almost instant and that the suction and Grasper of the Dobot are working on different material.

After testing each branch separately, we also run a few tests in the final combined version to make sure that there are no clashes happening and that everything runs in harmony.

7 Results:

Our integrated system successfully demonstrated robust real-time interaction capabilities, accurately detecting and interpreting gestures, colours, and objects under various lighting and environmental conditions. Through extensive testing, we achieved low-latency communication with the Dobot robotic arm, effectively controlling its actions based on visual and spoken user inputs. Gesture and colour detection maintained high accuracy (>90%) under moderate variations in lighting, while the Gemini LLM facilitated seamless and intelligent speech-based command processing. The GUI was effective in providing clear visual feedback and smooth operational control.

8 Project Management:

We managed our Project using Agile methodology in order to ensure flexibility of the project management. This approach works very well with the nature of the project where modifications were required at every stage of development.

8.1 Agile Methodology

The iterative nature of the Agile methodology is its fundamental component. Agile helped use to break our project into smaller tasks and make it easy to check that every part of our project is working fine. We always check that the project is going on the right track by checking our progress regularly and always reevaluating our work.

Agile methodology allows us to communicate well by sharing our progress and sharing the obstacles that faced every one of us during the working process and that made resolving problems in an easier way and be ready for the next stages.

In conclusion, using Agile was the best choice for us to achieve our goals in an easy way and maintaining a very good vision throughout the whole process, moreover, sharing our experience to achieve the maximum learning from our project.

8.2 Gantt Chart

Doing a Gantt Chart to help us monitor our overall progress by assigning tasks and deadlines and in contribution with the Agile methodology really helped our team to achieve our goals in a very easy and systematic way.

In the following figure are all the tasks throughout the term and their expected durations, this helped us to visualize the project timeline and to be ready for any unexpected situations that could prevent us from reaching our goal.

8.3 GitLab for Code Management

Throughout the project, we used GitLab as our code repository to help us see the contribution of each one in the project and to make it as easy as possible to mix all our work together, and to ensure coherency in our work.

Gitlab offers many tools that helped is

In order to make sure that new commits didn't cause mistakes, GitLab also offered useful tools for continuous integration (CI), where code changes were automatically tested. We were able to immediately correlate tasks and bugs to particular changes by using GitLab's issue tracking feature, which made it simpler to monitor progress and effectively address problems.

We made sure that our project stayed well-coordinated, adaptable, and organized by utilizing GitLab in conjunction with Agile and the Gantt chart. This allowed us to successfully complete the project on schedule.

9 Conclusion:

This project demonstrated a robust approach for an interactive teleoperation interface, successfully integrating speech, gesture, and visual interactions with minimal latency and high accuracy. Through careful optimization, particularly in image preprocessing and gesture recognition, the system-maintained responsiveness, proving suitable for practical human-robot communication applications. Its design provides a foundation for further advancements in seamless human-machine interactions.

9.1 Limitations:

Despite significant optimizations, latency remains a challenge, primarily due to computational intensity in real-time video stream processing and simultaneous multi-threaded tasks. Limited computational resources occasionally introduced delays during continuous video processing, particularly affecting real-time gesture and object detection speed. Although manageable, these occasional latency issues could impact user experience in highly dynamic situations.

9.2 Future Improvements:

To further enhance the system, future developments should focus on addressing latency by employing advanced techniques such as edge computing or hardware acceleration (e.g., dedicated GPUs or neural processing units). Implementing real-time optimization algorithms and exploring more efficient neural network models, like lighter-weight versions of YOLO or custom neural architectures tailored specifically for gesture and object detection, could significantly improve processing speeds. Additionally, optimizing the threading architecture to better balance workload distribution could reduce processing overhead, ensuring consistently fast interactions and improved overall responsiveness.

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